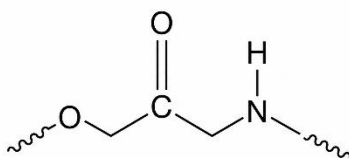


linkages could reduce activity depending on design context (Migawa et al., 2021). The main advantage of this chemistry is therefore not simply greater nuclease resistance, but a broader therapeutic-index gain: MsPA enables medicinal chemists to modulate the balance between stability, RNase H activity, protein binding, tissue exposure and tolerability more precisely than with uniformly PS-modified ASOs.

Backbone chemistries in combination with sugar modifications such as 2'-O-methoxyethyl (MOE) and locked nucleic acid (LNA), provide tunable levers for ASO pharmacology. Rather than simply replacing earlier chemistries, newer modifications expand the combinatorial design space available for therapeutic optimization. Modern oligonucleotide therapeutics increasingly use hybrid architectures that integrate multiple backbone and sugar modifications within a single molecule to balance nuclease stability, pharmacokinetics, potency, safety, and mechanism of action.

Morpholino Phosphorodiamidate Linkages

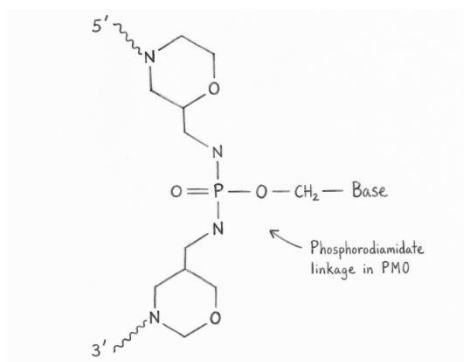


Carbamates are versatile amide–ester hybrid functional groups widely used in medicinal chemistry because they combine chemical stability, tunable conformational properties, and favorable pharmacokinetics. They serve as peptide bond surrogates, prodrug linkers, and key structural motifs in many approved drugs, enabling improved membrane permeability, metabolic stability, and controlled drug release. Their ability to modulate hydrogen bonding and resist enzymatic degradation has made them especially valuable in designing enzyme inhibitors and prodrugs across diverse therapeutic areas. In antisense technologies, carbamates are used to enhance stability and modulating molecular properties for delivery.

Carbamate-based therapeutics exhibit pharmacokinetic and pharmacodynamic profiles that are strongly influenced by their chemical stability and susceptibility to enzymatic cleavage. The carbamate-containing HIV protease inhibitor Darunavir is well absorbed orally with rapid attainment of peak plasma concentrations, and undergoes carbamate hydrolysis-dominated metabolism. PK can be significantly modulated by

co-administered agents; for example, ritonavir inhibits CYP3A metabolism, leading to markedly increased systemic exposure and prolonged half-life of darunavir. Excretion occurs via the feces, with minimal renal clearance, reflecting hepatic metabolism and biliary elimination. From a PD perspective, carbamates can either act as stable structural motifs that enhance target binding or as cleavable groups that enable controlled release of active species, thereby linking metabolic processing directly to therapeutic activity. Overall, carbamate functionalities provide a tunable balance between stability and bioactivation, which is central to optimizing both drug exposure and efficacy.

The development of phosphorodiamidate linkages (carbamate-like linkages) for morpholino antisense was a natural extension of the broader medicinal chemistry principles. Early antisense oligonucleotides were limited by the instability and charged nature of the phosphodiester backbone, prompting efforts to introduce chemically modified linkages that could enhance durability and biological performance. Building on small-molecule success of this carbamate chemistry, Dr James Summerton and colleagues introduced the phosphorodiamidate linkage as part of the morpholino platform. This modification replaced the negatively charged phosphodiester with a neutral backbone incorporating P–N bonds, conceptually analogous to carbamate functionality, yielding molecules that were highly resistant to nucleases while maintaining strong and specific hybridisation to RNA.



As this chemistry matured, phosphorodiamidate morpholino oligomers (PMOs) emerged as a distinct class of antisense agents that operate through steric blockade rather than enzymatic degradation, marking a shift toward safer and more controllable mechanisms of action (Summerton and Weller, 1997; Summerton, 2017). The introduction of phosphorodiamidates was a key translational step in antisense design.

Backbone modification is no longer used simply to increase nuclease resistance, but to fine-tune the broader pharmacological behavior of oligonucleotide drugs. When combined with sugar modifications such as MOE, LNA, or other affinity-enhancing groups, these backbone strategies provide a flexible design framework for balancing potency, stability, delivery, tissue exposure, safety, and mechanism of action. Next-generation antisense therapeutics are increasingly defined by rational hybrid architectures, in which each chemical modification is selected not as an isolated improvement, but as part of an integrated strategy to expand therapeutic index and clinical utility.